BEE : DC Circuits

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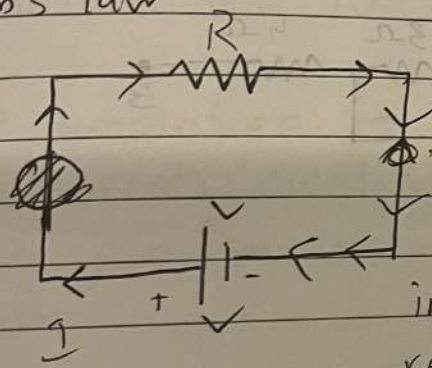
✯ Basics:

→ Resistance (R) : SI unit : Ω (OHM's)

$R = \rho \times \frac{l}{A}$ where ρ = specific resistivity
l = length of material
A = cross-sectional area

→ Capacitance (C) : SI unit : F (farad)
Acts as battery.

✯ Ohm's law

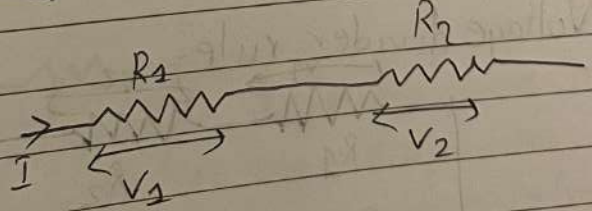


The current flowing in any closed circuit is directly proportional to the potential difference and inversely proportional to the resistance @ constant temp.

$\therefore I \propto V$ & $I \propto \frac{1}{R}$

$\therefore I = \frac{V}{R}$ $\therefore \boxed{V = IR}$

✯ Series resistance:



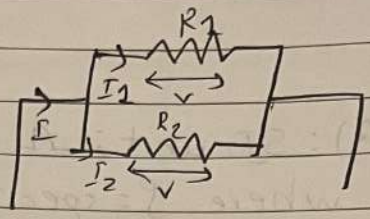
$R_{eq} = R_1 + R_2$

$V_1 = IR_1$; $V_2 = IR_2$

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★ Parallel resistance

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

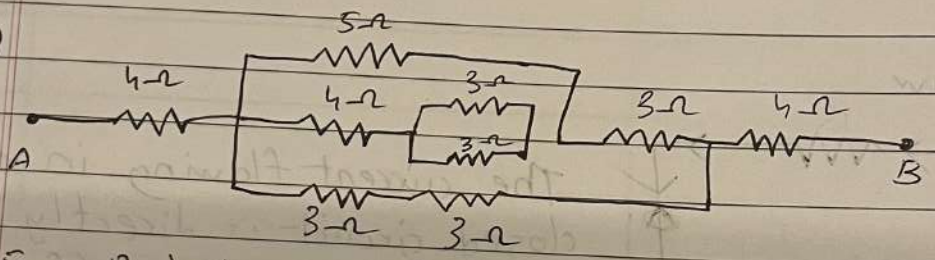


$V = I_1 R_1$; $V = I_2 R_2$

For 2 parallel resistors,

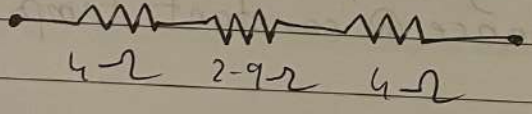
$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

eg)



Find R between A & B

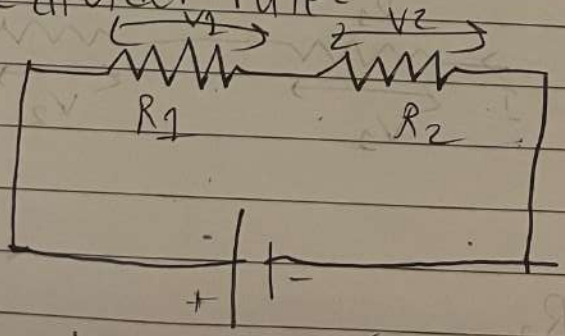
→ Simplified circuit:



$= 10.9\,\Omega$

★ Voltage divider rule:

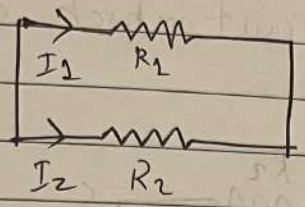
only applicable for series circuit.



$$V_1 = \left(\frac{R_1}{R_2 + R_1}\right) V_T \qquad V_2 = \left(\frac{R_2}{R_1 + R_2}\right) V_T$$

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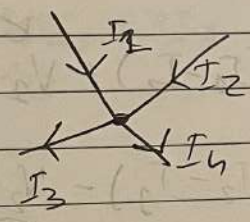
★ Current divider rule



Only for parallel circuits.

$$I_1 = \left(\frac{R_2}{R_1+R_2}\right) I_T \qquad I_2 = \left(\frac{R_1}{R_1+R_2}\right) I_T$$

★ Junction law: (KCL)

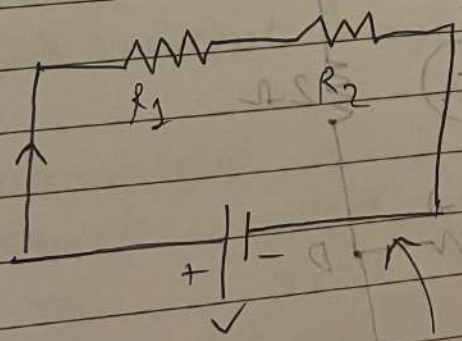


$\sum I_n = 0$

The summation of all currents meeting at a junction is zero

∴ For the above eg,
$I_1 + I_2 - I_3 - I_4 = 0$ (incoming currents = +ve, outgoing currents = -ve)

★ KVL:

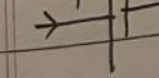


The summation of voltage drop across a closed circuit is always zero.

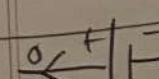
$\sum V = 0$

$-IR_1 - IR_2 + V = 0$

For resistances:



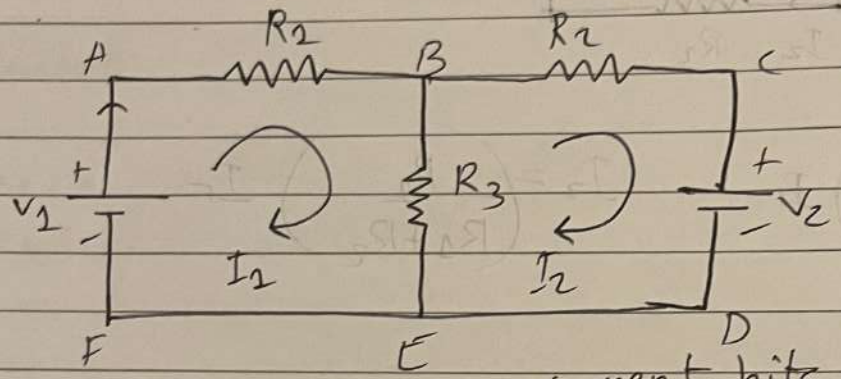
= -ve



= +ve

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★ Mesh analysis:
A mesh is a closed circuit which cannot further be divided.



current hits -ve terminal first

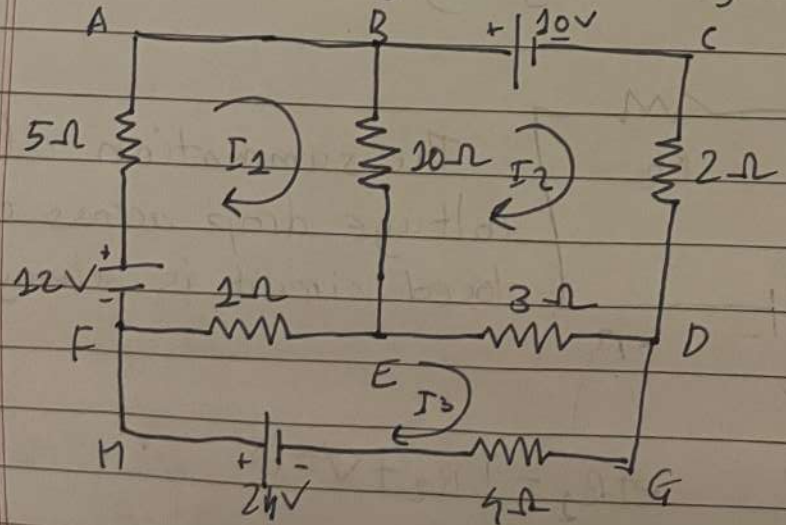
$-I_1R_1 - R_3(I_1 - I_2) + V_1 = 0$ ← mesh ABEFA

$-I_2R_2 - R_3(I_2 - I_1) - V_2 = 0$ ← mesh BCDEB

current hits +ve terminal first

→ Steps for mesh analysis:
i) Assume clockwise current
ii) Apply KVL to each loop.
iii) Solve eqns to get I_n.
iv) Find current through branch using required I_n.

eg)



→ ABEFA

$-5I_1 - 10(I_1 - I_2) - I_1 + 12 = 0$

$-16I_1 + 10I_2 = -12$

$-5I_1 - 10(I_1 - I_2) - (I_2 - I_3) + 12 = 0$

$-16I_1 + 10I_2 + I_3 = -12$

→ BCDEB

$-10(I_2 - I_1) - 3(I_2 - I_3) - 2I_2 - 10 = 0$

$-10I_2 - 3I_2 - 2I_2 + 10I_1 + 3I_3 = 10$

$-15I_2 + 10I_1 + 3I_3 = 10$

→ FDGHF

$-(I_3 - I_1) - 3(I_3 - I_2) - 4I_3 + 24 = 0$

$-8I_3 + I_1 + 3I_2 = -24$

$I_1 = \frac{1582}{901}$ $I_2 = \frac{1118}{901}$ $I_3 = \frac{3320}{901}$

$\therefore I_1 = 1.75$ $I_2 = 1.24$ $I_3 = 3.68$

$I_3 = 3.68$ = current through 4Ω

☆ Supermesh analysis:

Use when: When there's a current source shared by 2 meshes.

→ Steps:

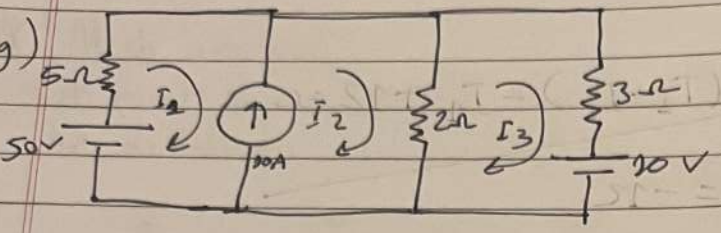
i) $I_1 - I_2$ = Shared current.

ii) Remove shared current, make a big mesh, apply KVL.

iii) Solve eqns.

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eg)



Find current through 5Ω.

→ ~~$I_1 - I_2 = 10A$~~ $I_2 - I_1 = 10A$

$\therefore I_2 = 10 + I_1$

~~$-5I_1 - 2(I_2 - I_3) - 3I_3 - 10 + 50 = 0$~~

~~$-5I_1 - 2I_2$~~

Apply KVL to supermesh:

$-5I_1 - 2(I_2 - I_3) + 50 = 0$

$-5I_1 - 2I_2 + 2I_3 = -50$

Apply KVL to third loop:

$-3I_3 - 2(I_3 - I_2) + 10 = 0$

$-5I_3 + 2I_2 = +10$

$\therefore I_1 = 5.4A$

$\therefore I_2 = 15.48A$

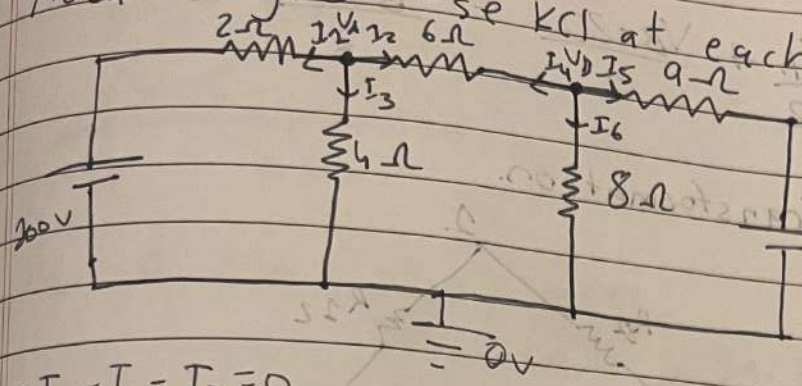
$\therefore I_3 = 4.19A$

$I_{5\Omega} = I_1 = 5.4A$

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★ Nodal analysis: Use KCl at each node.

eg)



Find current through 6Ω.

$-I_1 - I_2 - I_3 = 0$

$\therefore I_1 + I_2 + I_3 = 0 \qquad I_4 + I_5 + I_6 = 0$

$$\frac{V_A - 200}{2} + \frac{V_A}{4} + \frac{V_A - V_B}{6} = 0$$

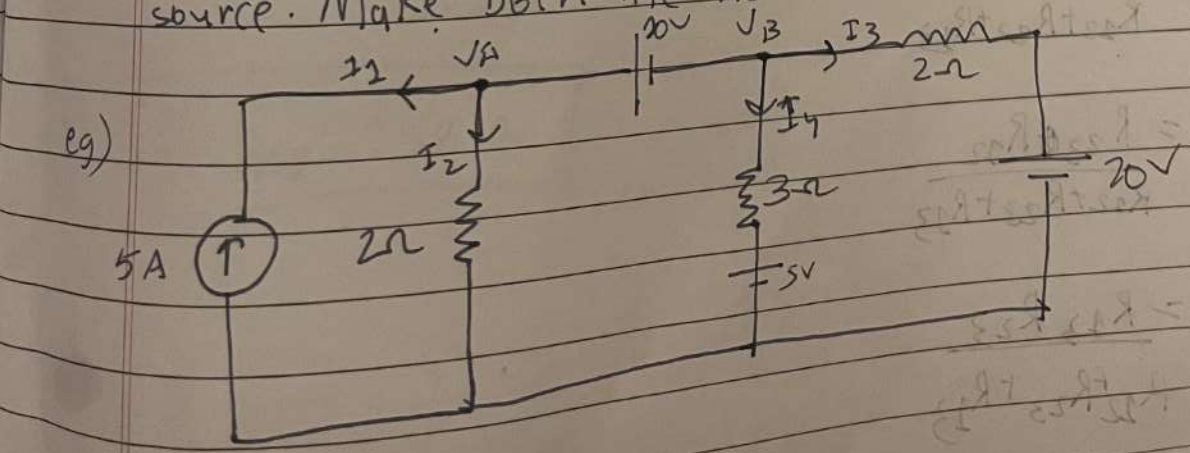
$$\frac{V_B - V_A}{6} + \frac{V_B - 20}{9} + \frac{V_B}{8} = 0$$

$V_A = 60.06$, $V_B = 30.36$

$$I_3 = I_{6\Omega} = \frac{60.06 - 30.36}{6} = 4.95A$$

★ Supernode analysis: When 2 nodes share a voltage source. Make both the nodes one node.

eg)

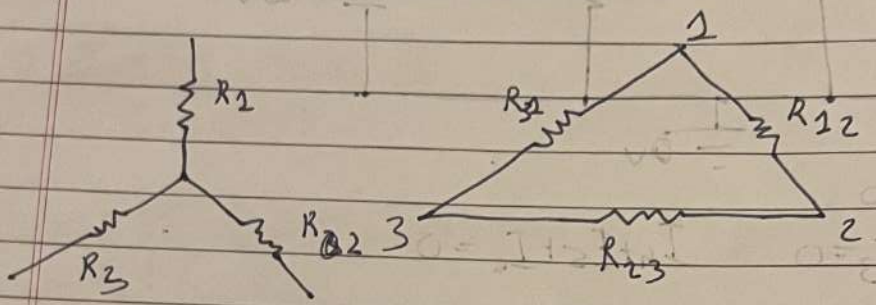


$V_A - V_B = 10$

$I_1 + I_2 + I_3 + I_4 = 0$

$$-5 + \frac{V_A}{2} + \frac{V_B - 20}{2} + \frac{V_B - 5}{3} = 0$$

☆ Star-Delta Transformation.



Star

Delta

→ To convert from star to delta

$$R_{12} = R_1 + R_2 + \frac{R_1 R_2}{R_3}$$

$$R_{23} = R_2 + R_3 + \frac{R_2 R_3}{R_1}$$

$$R_{13} = R_1 + R_2 + \frac{R_1 R_3}{R_2}$$

→ To convert from delta to star

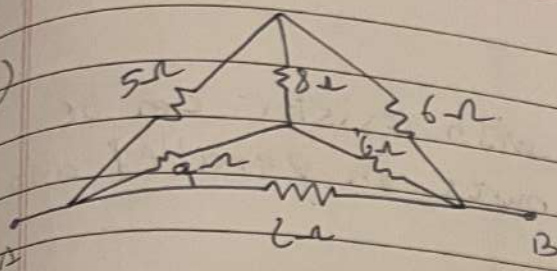
$$R_1 = \frac{R_{12} R_{13}}{R_{12} + R_{23} + R_{13}}$$

$$R_2 = \frac{R_{23} R_{12}}{R_{12} + R_{23} + R_{13}}$$

$$R_3 = \frac{R_{13} R_{23}}{R_{12} + R_{23} + R_{13}}$$

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e1)

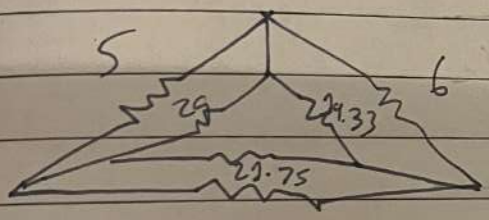


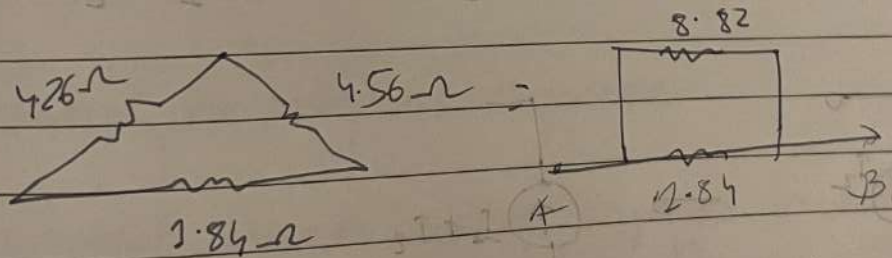
. Find R_{AB}.

$R_{12} = 8 + 6 + \frac{48}{9} = 19.33\ \Omega$

$R_{23} = 15 + \frac{54}{8} = 21.75\ \Omega$

$R_{13} = 17 + \frac{72}{6} = 29\ \Omega$





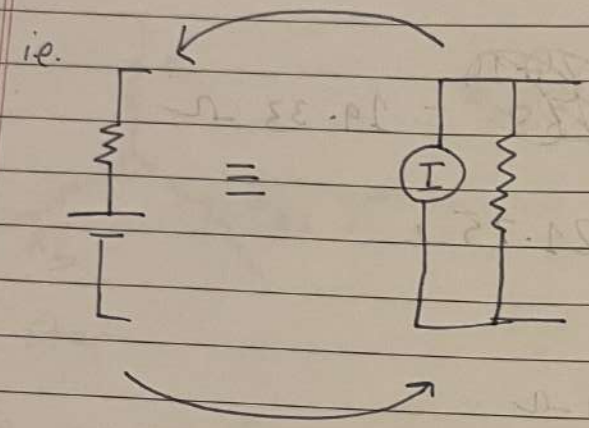
$\therefore R_{AB} = 1.52\ \Omega$

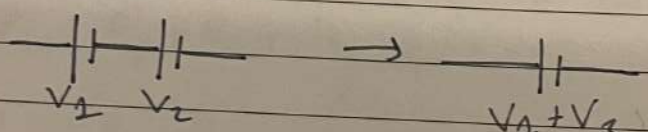
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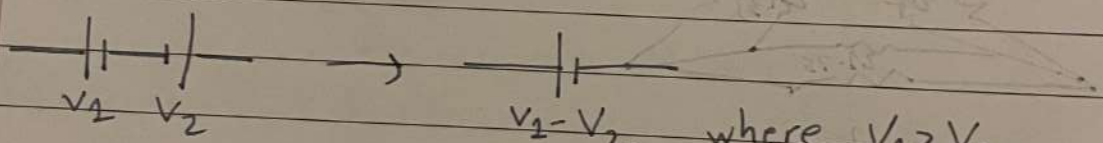
✡ Source Transformation:

Voltage source in series with resistor can be converted to current source in parallel and vice versa.

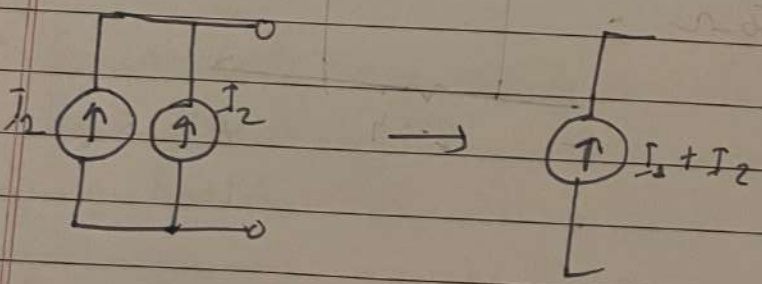
ie.

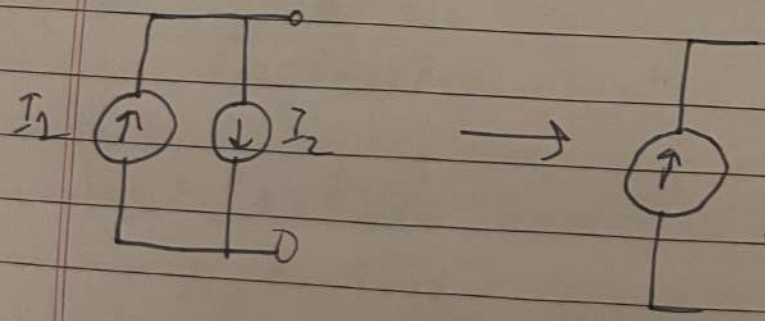






where $V_1 > V_2$

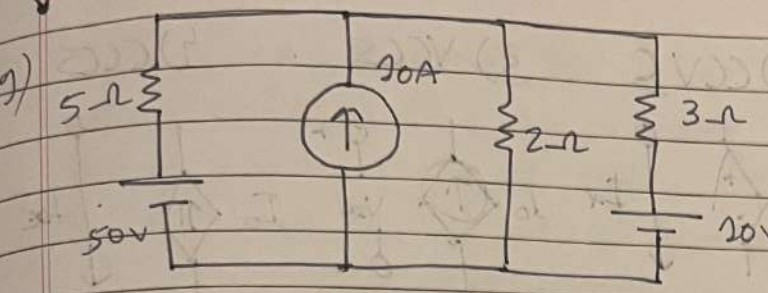




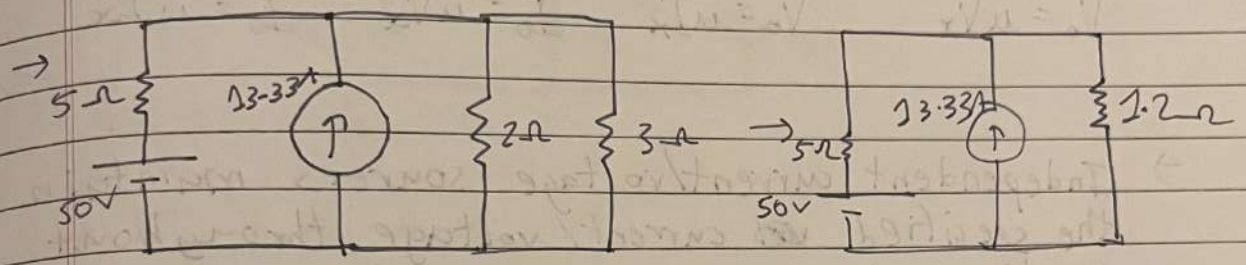
where $I_1 > I_2$

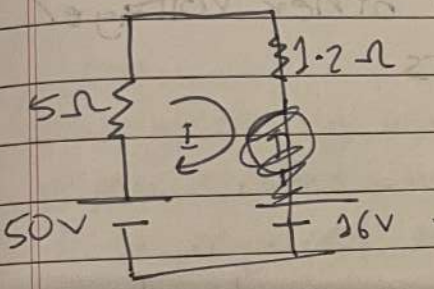
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eg)



Find I through 5Ω resistor.





$-5I - 1.2I - 16 + 50 = 0$

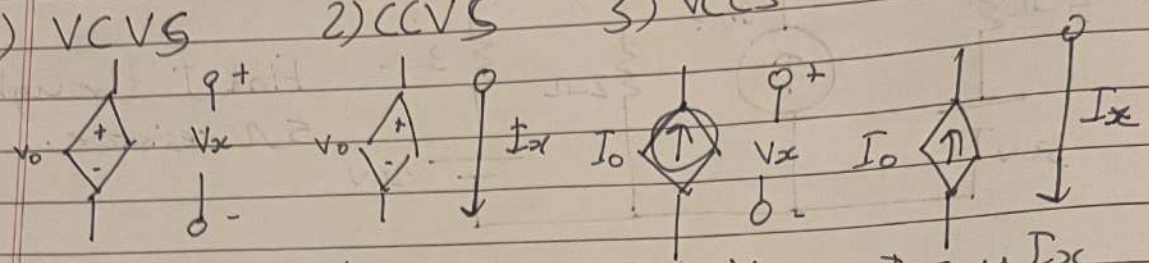
$\boxed{I = 5.18A}$

✡ Theory:

1) Dependent vs. independent sources.

→ Dependent sources' value depends on other voltage or current somewhere else in the circuit. Its value is controlled by a circuit variable. Hence, they are also called controlled sources. There are dependent voltage sources and dependent current sources.
The types of dependent sources are:

1) VCVS 2) CCVS 3) VCCS 4) CCCS



$V_o = \mu V_x$ $V_o = \mu I_x$ $I_o = \mu V_x$ $I_o = \mu I_x$

→ Independent current/voltage sources maintain the specified current/voltage throughout its terminals, regardless of other voltage/current or other components.

★ Ideal vs Practical sources

1) Ideal voltage sources maintain a constant voltage across its terminals regardless of current A practical voltage source has a small internal resistance.

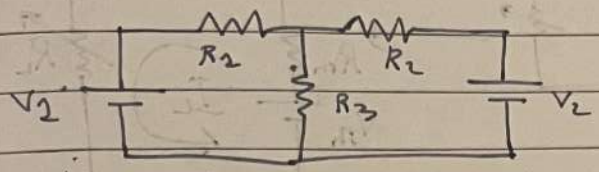
$$V_L = V_o - I R_o$$

V_L: load voltage; V_o: ideal source's voltage; I: load current; R_o: internal resistance

2) Ideal current sources supply a constant current regardless of voltage. Practical current sources have a large but finite resistance when connected in parallel with the ideal current source.

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☆ Superposition theorem: The total current flowing in any resistance due to all the sources is equal to the algebraic sum of the current flowing in a circuit due to individual sources.



V_1 V_2 V_1 V_2

$I_T = I_1 + I_2$

→ Steps:

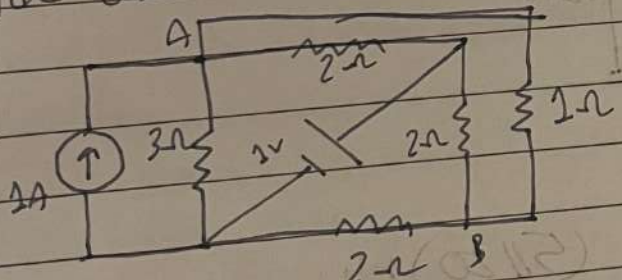
i) Consider only one source and short circuit other voltage sources. Open circuit current sources.

ii) Find required current.

iii) Repeat.

iv) Add obtained current.

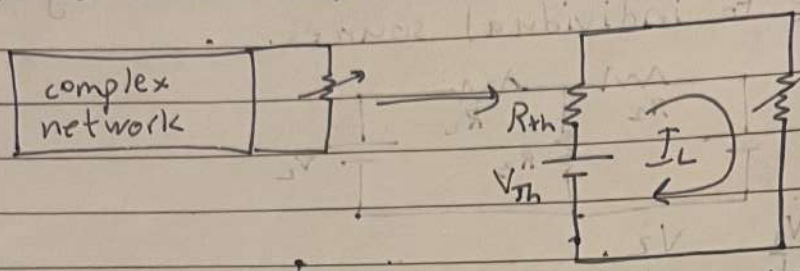
eg)



apply KCL at A and B,
solve later ☺

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☆ Thevenin's theorem: Any complex network can be converted into single voltage source (V_{Th}) in series with R_{Th}



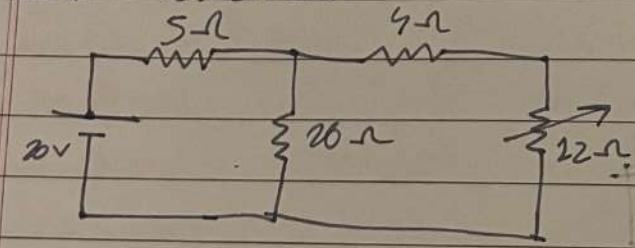
& $V_L = I_L \times R_L$

where $I_L = \frac{V_{th}}{R_{th} + R_L}$

→ Steps:

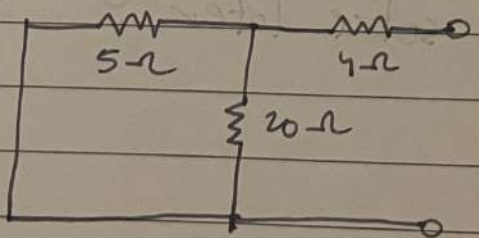
1) To find V_{th}, remove R_L from circuit & find V_{th} using mesh analysis.
2) To find R_{th}, short circuit voltage; open circuit current; solve resistance.

eg)



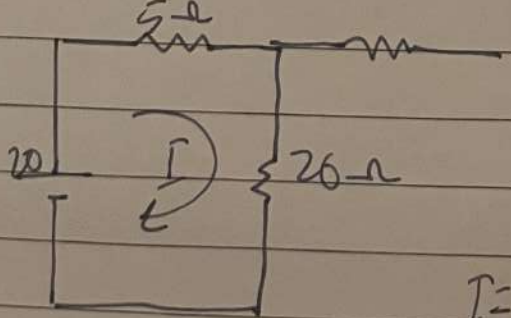
Find current through R_L

→ for R_{th}



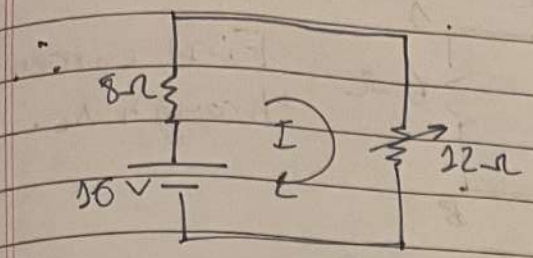
$(5 \| 20) + 4$
$= 8\Omega = R_{th}$

for V_{th}



$20 - 5I - 20I = 0$
$20 = 5I + 20I$
$I = 0.8A$

$V = 20 \times 0.8$
$= 16V$

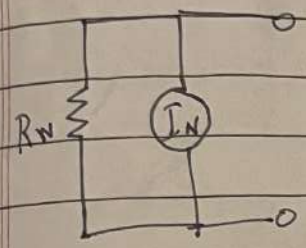


$-8I - 12I = -16$

$8I + 12I = 16$

$I = 0.8A$

★ Norton's theorem: Any linear two-terminal network containing independent sources and resistors can be simplified to a current source in parallel with a resistance.



Norton equiv.

$$I_L = \frac{I_N \times R_N}{R_N + R_L}$$

→ Thevenin & Norton's relation:

i) $R_N = R_{Th}$

ii) $I_N = \frac{V_{Th}}{R_{Th}}$

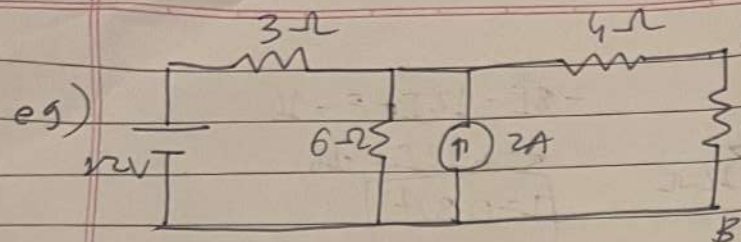
iii) $V_{Th} = I_N \times R_N$

→ Steps:

1) Remove R_L and short circuit it to find I_N.
2) Remove R_L and open circuit it to find R_N.
3) Load current $= \frac{I_N R_N}{R_N + R_L}$

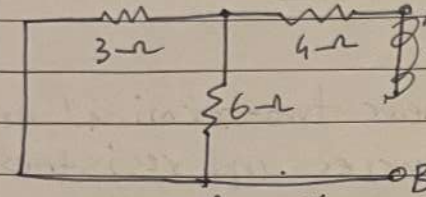
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eg)



Find current through R_L.

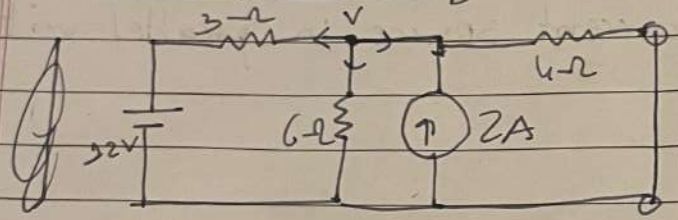
→



← To find R_N

$(3 \| 6) + 4$

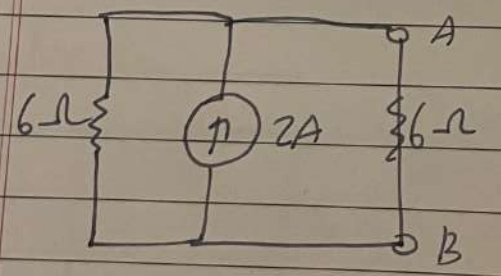
$= 6\,\Omega = R_N$



$$\frac{V_0 - 12}{3} + \frac{V}{6} - 2 + \frac{V}{4} = 0$$

$V = 8V$

$I_N = \frac{V}{R_{eq}} = 2A$



$I_N =$ ~~2~~ ×

$$I_L = \frac{6 \times 2}{12} = 1A$$

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★ Maximum Power Transfer Theorem: A load receives max. power (I^2R) from a linear network when $R_L = R_{TH} = R_N$.

→ Derivation:

$P = I^2 R_L$ ← (1)

$I = \frac{V_{TH}}{(R_{TH} + R_L)}$ ← (2)

Putting (2) in (1):

$P = \frac{V_{TH}^2 R_L}{(R_{TH} + R_L)^2}$

$\frac{dP}{dR_L} = \frac{(R_{TH} + R_L)^2 - 2R_L(R_{TH} + R_L)}{} = 0$ (for max.)

Rearranging,

$\therefore R_{TH} + R_L = 2R_L$

$\therefore R_{Th} = R_L$

→ if: $R_L > R_{Th}$ efficiency > 50% (less than max.)
$R_L < R_{Th}$ efficiency < 50%. (less than max.)

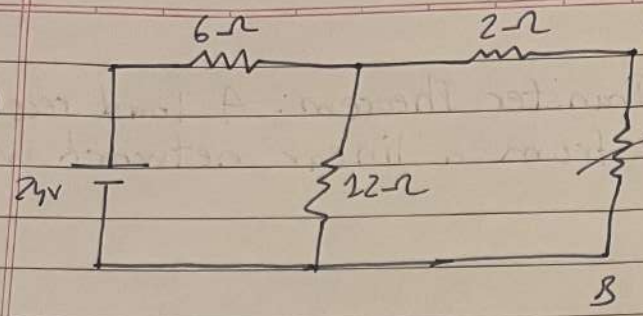
Power to load ↓

→ Steps:

1) Remove R_L between A & B
2) Find R_{Th} by removing sources.
3) Find V_{Th} by open circuiting A & B.
4) Set $R_L = R_{Th}$
5) $P = \frac{V_{Th}^2}{(4R_{Th})}$

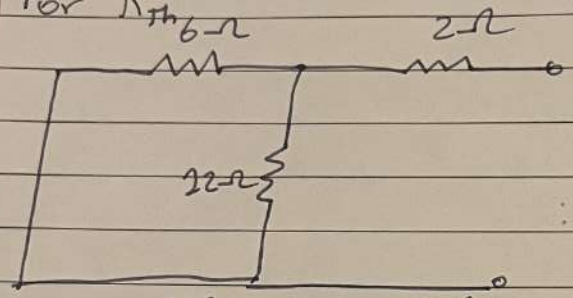
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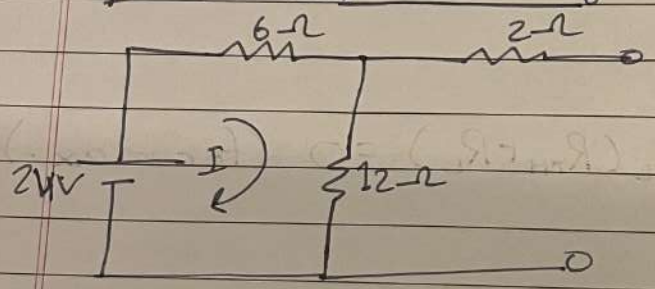
Find R_L for max. power transfer and max. power.

→ For R_{Th}



$(6 // 12) + 2$

$R_{Th} = 6\Omega$



$-6I - 12I + 24 = 0$

$6I + 12I = 24$

$I + 2I = 4$

$I = 1.75A$

$\therefore V_{Th} = 12 \times 1.75$

$= 16V$

$\therefore R_L = 6\Omega$

& $P_{max} = \frac{16^2}{24} = 10.67W$